



2011
TRIAL
HIGHER SCHOOL CERTIFICATE

GIRRAWEEN HIGH SCHOOL

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board – approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

Attempt Questions 1 – 10

All questions are of equal value

Question 1 (12 marks)**Marks**

- (a) Simplify $2x^2 + 3x - x(2x - 1)$ **2**
- (b) Solve for x :
- (i) $|2x - 3| \leq 7$ **2**
- (ii) $\frac{x}{3} + \frac{2x-1}{4} = 24$ **2**
- (c) Find the exact value of a and b if $a + b\sqrt{2} = \frac{3}{2\sqrt{2}-1}$ **2**
- (d) Solve for x : $2x^2 - x - 3 = 0$ **2**
- (e) Find the value of $\frac{\pi^2+1}{\sqrt{2}-1}$. Answer correct to three significant figures. **2**

Question 2 (12 marks)

- (a) Differentiate with respect to x :
- (i) $x^2 e^{5x}$ **2**
- (ii) $\frac{\sin x}{x^2+1}$ **2**
- (iii) $(\ln x)^2$ **2**
- (b) Find
- (i) $\int x^3 - 4x + 1 \cdot dx$ **2**
- (ii) $\int_0^{\frac{\pi}{12}} \cos 3x \cdot dx$ **2**
- (iii) $\int \sqrt{3x-1} \cdot dx$ **2**

Examination continues on the next page

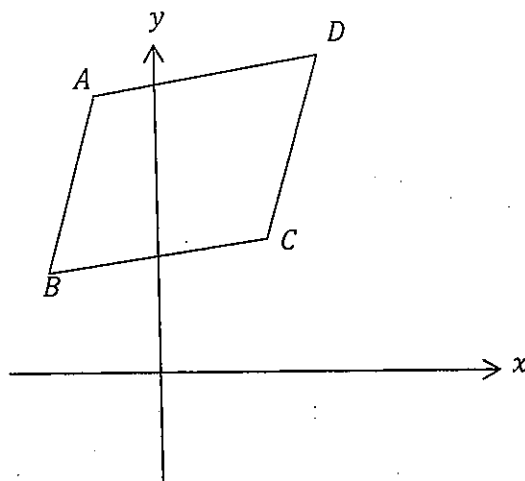
Question 3 (12 Marks)

Marks

- (a) Solve for θ , $0 \leq \theta \leq 2\pi$: $2 \sin \theta = \sqrt{3}$
- (b) A quadrilateral is formed with vertices $A(-1,4)$, $B(-2,1)$, $C(1,2)$ and $D(2,5)$ (See below)

2

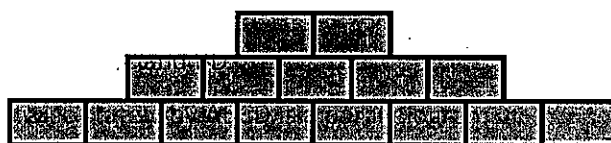
DIAGRAM NOT TO SCALE



- | | | |
|-------|---|----------|
| (i) | Find the midpoints of AC and BD. | 2 |
| (ii) | Hence or otherwise state why ABCD is a parallelogram. | 1 |
| (iii) | Find the gradients of AC and BD. | 2 |
| (iv) | Hence or otherwise state why ABCD is a rhombus. | 1 |
| (v) | Find the lengths of AC and BD. | 2 |
| (vi) | Hence or otherwise find the area of rhombus ABCD. | 2 |

Question 4 (12 marks)

- (a) A brick monument is to be built as below, with 2 bricks in the top row, 5 in the second top row, 8 in the third row and so on:



- | | | |
|------|--|----------|
| (i) | How many bricks will it take to fill 20 rows? | 2 |
| (ii) | There are 1000 bricks available. How many complete rows can be made? | 3 |

Question 4 continues on the next page

(b) $ABCD$ is a square. $AP = BQ$ and $\angle PQR = 90^\circ$.

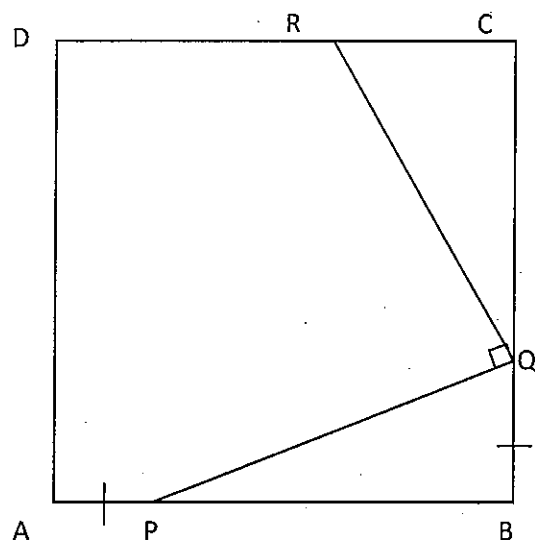


DIAGRAM NOT TO SCALE

(i) Prove $\triangle PBQ \equiv \triangle QCR$

2

(ii) Hence prove $AP = CR$.

1

(c) $ABCD$ is a quadrilateral with $AB = CD$ and $\angle ABC = \angle DCB$.

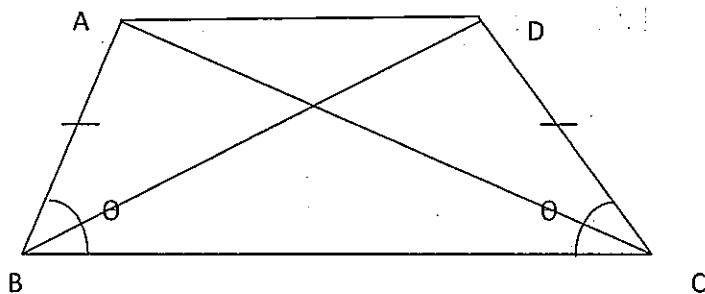


DIAGRAM NOT TO SCALE

(i) Prove $\triangle ABC \equiv \triangle DCB$.

1

(ii) Hence OR otherwise prove $\triangle BAD \equiv \triangle CDA$.

1

(iii) Hence or otherwise prove $ABCD$ is a trapezium.

2

Examination continues on the next page

Question 5 (12 Marks)**Marks**

- (a) Find the equation of the tangent to the curve $y = e^{2x}$ at the point where $x = 1$. **3**
- (b) For the curve $f(x) = 3x^2 - x^3$
- (i) Find the co-ordinates of the points of intersection with the x and y axes. **2**
- (ii) Find the co-ordinates of the stationary points of $f(x) = 3x^2 - x^3$ and determine their nature. **4**
- (iii) Find the point of inflexion of $f(x) = 3x^2 - x^3$ **1**
- (iv) Sketch the graph of $f(x) = 3x^2 - x^3$ showing all stationary points, points of inflexion and intercepts with the x and y axes. **2**

Question 6 (12 Marks)

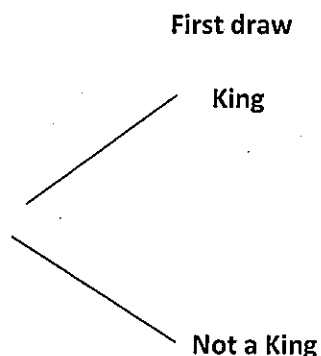
- (a) Solve for x where x is a real number: $e^{2x} - 2e^x - 3 = 0$ **3**
- (b) Find the equation of $f(x)$ if $f'(x) = 6 \cos 2x$ and $f(x) = \frac{3\sqrt{2}}{2}$ when $x = \frac{5\pi}{8}$. **2**
- (c) A particle is moving along a straight line so that its velocity at any time t is given by $v = t^2 - 4t$. If the particle is initially four metres to the right of $x = 0$:
- (i) Find the expression for acceleration in terms of time. **1**
- (ii) Find the minimum velocity of the particle. **2**
- (iii) Find an expression for the displacement x in terms of time. **2**
- (iv) Find the total *distance* the particle travels in the first five seconds. **2**

Examination continues on the next page

Question 7 (12 marks)**Marks**

- (a) From a standard deck of cards (4 cards of each kind, 13 of each suite 52 cards in total), 3 cards are drawn without replacement.

- (i) Complete the probability tree below on your own paper displaying the probability of drawing kings from a standard deck without replacement:

2

- (ii) Find the probability of drawing 2 kings in the 3 draws. **2**
- (b) Use Simpson's Rule with 4 subintervals to find an approximation for $\int_0^2 3^x \cdot dx$ **2**
- (c) 100 goats were released on to an island at the start of 1850 to provide food for shipwrecked sailors. By 1855 there were 180 goats on the island. Assuming the goat population since 1850 to follow the rule $P = Ae^{kt}$ where t is the time in years since 1850:
- (i) Find the values of A and k . **3**
- (ii) Find the expected goat population in 1860. **1**
- (iii) Once there are more than 1000 goats on the island they start to eat more vegetation than the island can regenerate. In what year did this occur? **2**

Examination continues on the next page

Question 8 (12 Marks)

Marks

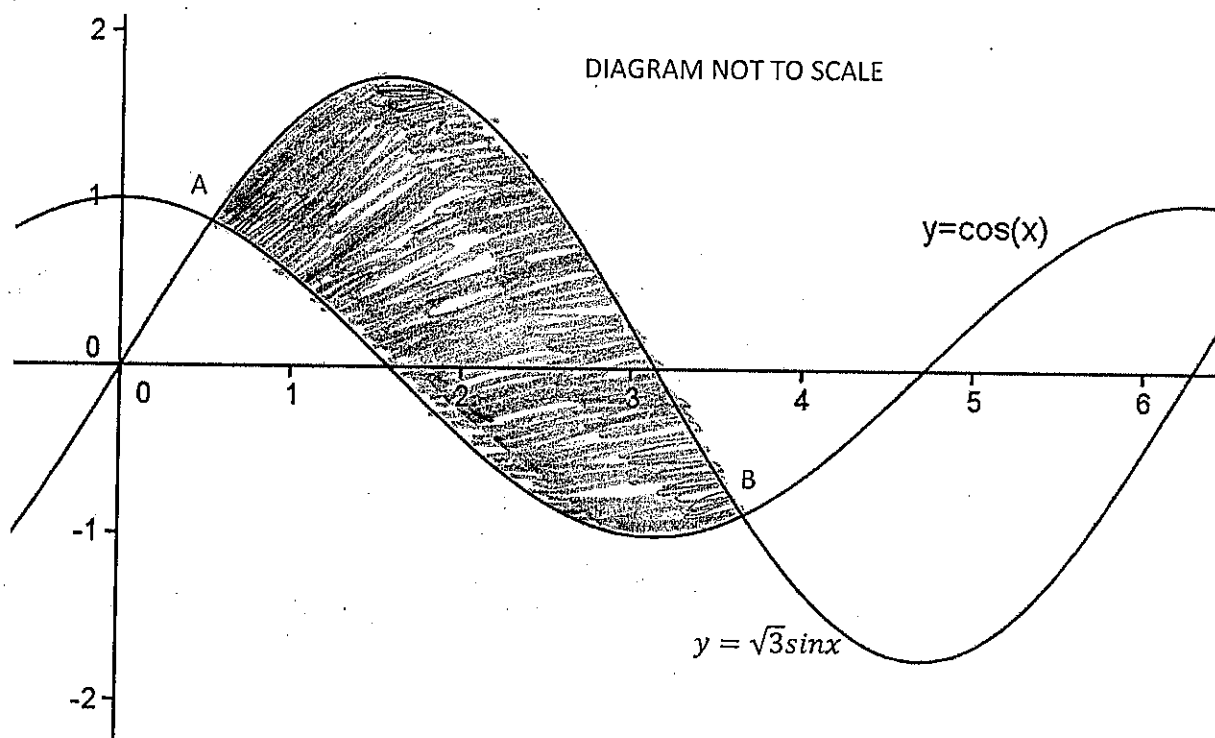
(a) (i) Solve the equation $\cos x = \sqrt{3} \sin x$ for $0 \leq x \leq 2\pi$.

3

(ii) Hence find the shaded area between the points A and B

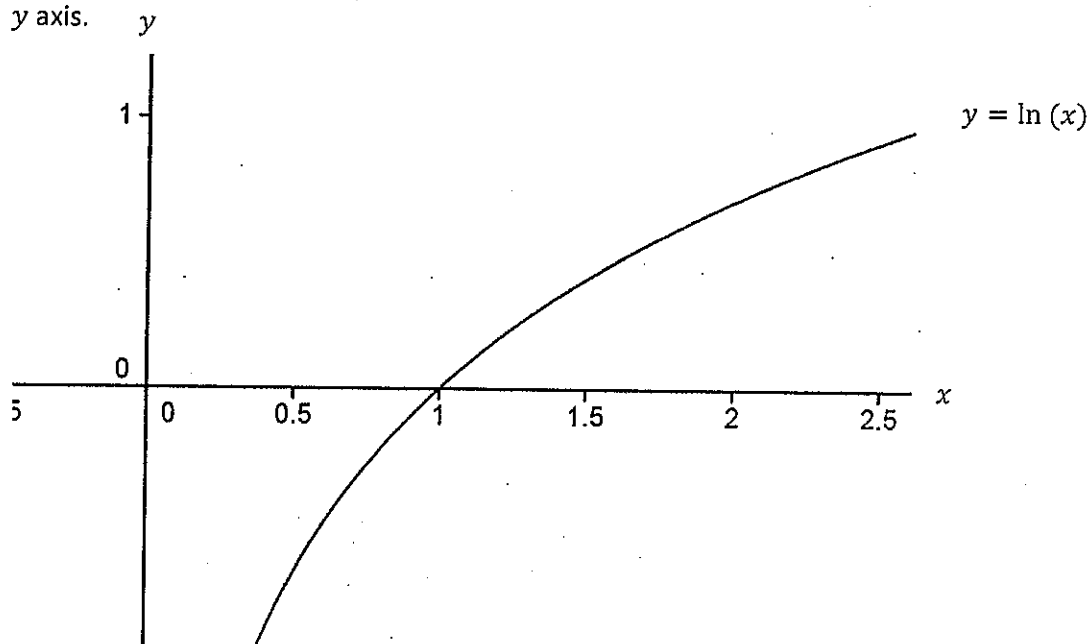
2

(the intersection points of $y = \cos x$ and $y = \sqrt{3} \sin x$)



(b) Find the volume of the solid of revolution formed when the area between the curve $y = \ln(x)$, the y axis and the lines $y = 0$ and $y = 1$ is rotated around the y axis.

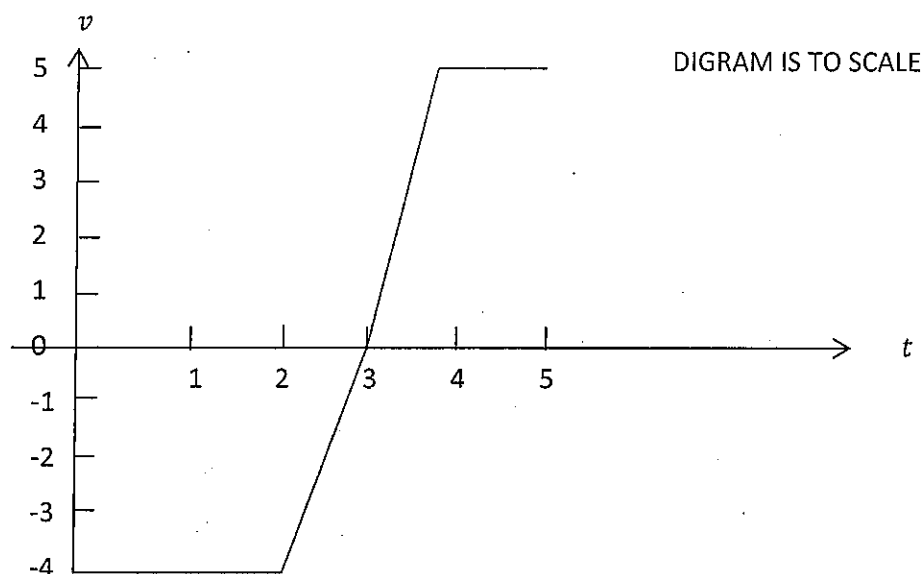
4



Question 8 (continued)

Marks

(c) The graph below shows the velocity of a particle at time t seconds:



If the particle started 2 metres to the right of $x = 0$ where did it end up after 5 seconds?

3

Question 9 (12 marks)

(a) Rebecca invests \$8000 per year in a superannuation fund which pays 9%P.A. interest.

(i) Show that the amount Rebecca has in her fund after 3 years

2

$$\text{Is } \frac{8000 \times 1.09(1.09^3 - 1)}{0.09}$$

(ii) How much will Rebecca have in her superannuation fund after 20 years?

1

(iii) At the rate at which Rebecca is investing, how many years will it take for her superannuation fund to be worth \$1000 000?

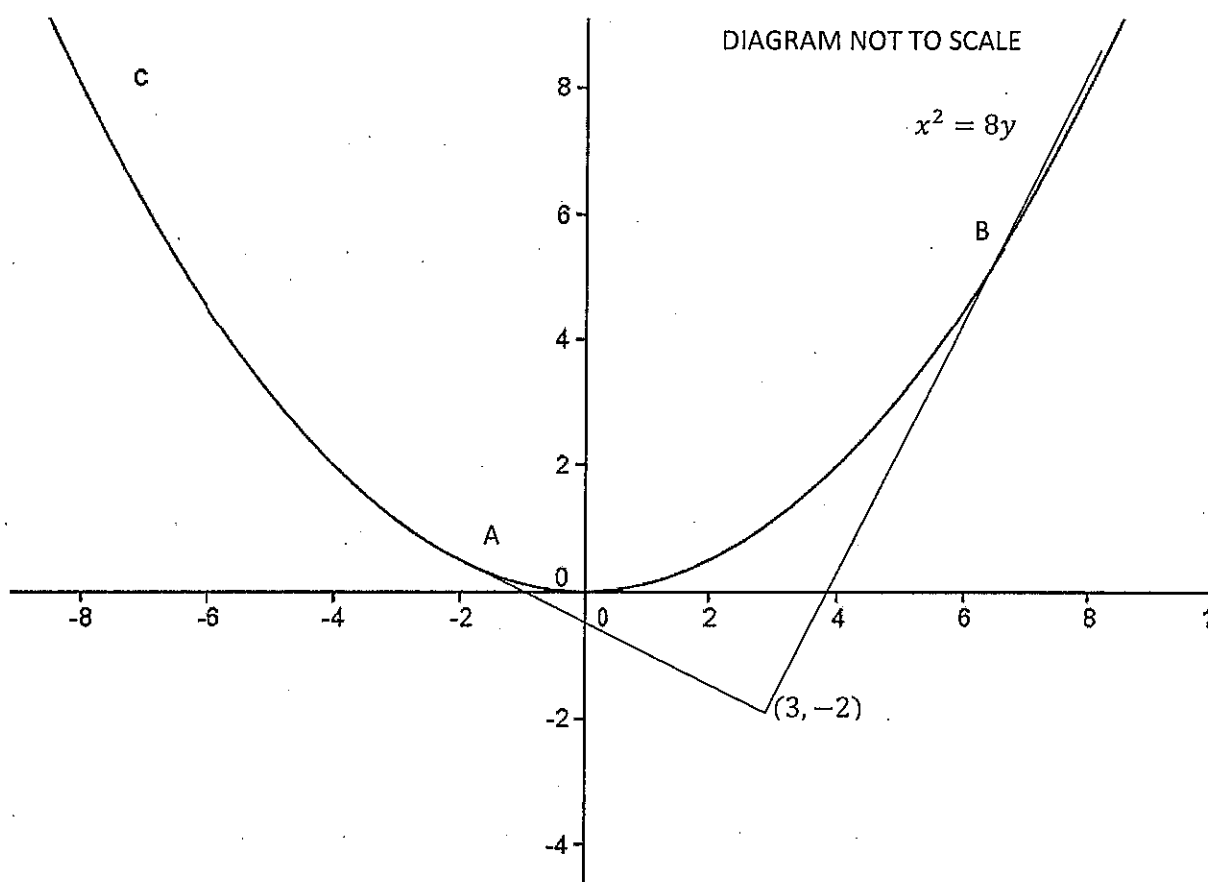
2

Examination continues on the next page

Question 9 (continued)

Marks

- (b) (i) Show that the equation of a line through the point $(3, -2)$ with gradient m is given by $mx - y - 3m - 2 = 0$. 1
- (ii) Show that if a line through $(3, -2)$ with gradient m is a tangent to the parabola $x^2 = 8y$ then $64m^2 - 96m - 64 = 0$. 3
- (iii) Find the two points A and B on $x^2 = 8y$ where the tangents from $(3, -2)$ meet the parabola $x^2 = 8y$. 2
- (iv) Show that AB is a focal chord. 1



Examination continues on the next page

Question 10 (12 marks)

Marks

- (a) Haasika is paying off a home loan of \$500 000 over 20 years in equal monthly instalments at a rate of 7.2%P.A. If she is paying off the loan at \$P per month:

- (i) Show that the amount she has left to repay after 3 months is given by 1

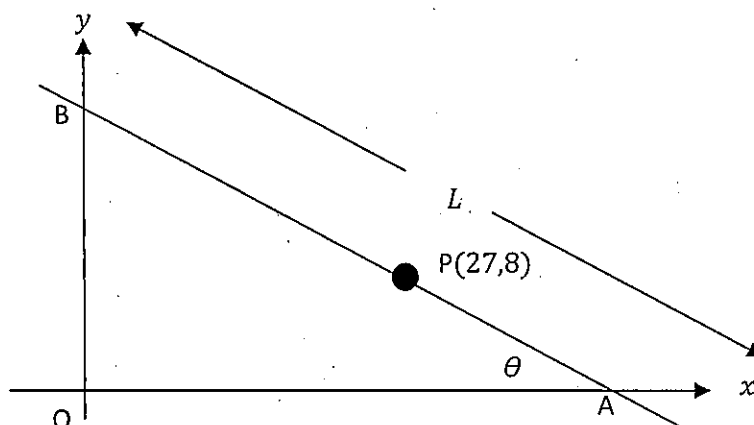
$$A = \$500\,000 \times 1.006^3 - P(1 + 1.006 + 1.006^2)$$

- (ii) Find an expression for the amount Haasika has left to repay after n months. 2

- (iii) Find the amount of each monthly repayment. 1

Answer correct to the nearest dollar.

- (b) In the diagram below, a straight line passes through the fixed point P (27,8) and meets the x and y axes at A and B respectively. L is the distance along the line AB and the acute angle $\angle OAB = \theta$.



- (i) Show that $L = \frac{27}{\cos \theta} + \frac{8}{\sin \theta}$ 2
- (ii) Show that $\frac{dL}{d\theta} = \frac{27\sin^3\theta - 8\cos^3\theta}{\sin^2\theta\cos^2\theta}$ 1
- (iii) Hence show that L is a minimum when $\tan \theta = \frac{2}{3}$. 3
- (iv) Find the minimum distance between A and B. 2

END OF EXAMINATION

$$\begin{aligned} \text{Q. (1)(a)} \quad 2x^2 + 3x - x(2x-1) \\ = 2x^2 + 3x - 2x^2 + x \\ = 4x \end{aligned}$$

$$\begin{aligned} \text{(b)(i)} \quad |2x-3| &\leq 7 \\ -7 &\leq 2x-3 \leq 7 \\ -4 &\leq 2x \leq 10 \\ -2 &\leq x \leq 5 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \frac{x}{3} + \frac{2x-1}{4} &= 24 \\ 4x + 6x - 3 &= 288 \\ 10x - 3 &= 288 \\ 10x &= 291 \\ x &= 29.1 \text{ or } 29 \frac{1}{10} \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad a+b\sqrt{2} &= \frac{3}{2\sqrt{2}-1} \times \frac{(2\sqrt{2}+1)}{(2\sqrt{2}+1)} \\ &= \frac{6\sqrt{2}+3}{8-1} \\ a+b\sqrt{2} &= \frac{6\sqrt{2}+3}{7} \end{aligned}$$

$$\therefore a = \frac{3}{7}, b = \frac{6}{7}$$

$$\begin{aligned} \text{(d)} \quad 2x^2 - x - 3 &= 0 \\ (2x-3)(x+1) &= 0 \\ x &= \frac{3}{2} \text{ or } x = -1 \\ \text{(e)} \quad \frac{\pi^2+1}{\sqrt{2}-1} &= \dots \\ &= 26.2415 \dots \\ &= 26.2 \text{ (to 3 significant figures)} \end{aligned}$$

$$\begin{aligned} \text{Q. (2)(a)(i)} \quad y &= x^2 e^{5x} \\ \frac{dy}{dx} &= u'v + v'u \end{aligned}$$

$$\begin{aligned} &= 2xe^{5x} + 5e^{5x} \times x^2 \\ &= xe^{5x}(2+5x) \end{aligned}$$

$$\text{(ii)} \quad y = \frac{\sin x}{x^2+1}$$

$$\frac{dy}{dx} = \frac{u'v - v'u}{v^2}$$

$$= \frac{\cos x(x^2+1) - 2x \sin x}{(x^2+1)^2}$$

$$\frac{dy}{dx} = \frac{x^2 \cos x + \cos x - 2x \sin x}{(x^2+1)^2}$$

$$\begin{aligned} \text{(iii)} \quad y &= [\ln(x)]^2 \quad \text{Letting } u = \ln x \\ \frac{dy}{dx} &= \frac{dy}{du} \times \frac{du}{dx} \quad \frac{du}{dx} = \frac{1}{x} \\ y &= u^2 \\ \frac{dy}{du} &= 2u \\ &= 2 \ln x \times \frac{1}{x} \end{aligned}$$

$$\frac{dy}{dx} = \frac{2 \ln x}{x}$$

$$\begin{aligned} \text{(b)(i)} \quad \int x^3 - 4x + 1 \, dx \\ = \frac{1}{4}x^4 - 2x^2 + x + C \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \int_0^{\frac{\pi}{2}} \cos 3x \, dx \\ = \left[\frac{1}{3} \sin 3x \right]_0^{\frac{\pi}{2}} \\ = \frac{1}{3} \sin \frac{\pi}{2} - \frac{1}{3} \sin 0 \\ = \frac{\sqrt{2}}{6} \end{aligned}$$

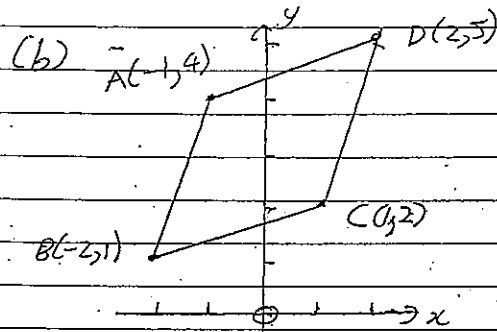
$$\begin{aligned} \text{(iii)} \quad \int \sqrt{3x-1} \, dx \\ = \frac{1}{3x-\frac{1}{2}} (3x-1)^{\frac{3}{2}} + C \end{aligned}$$

$$\begin{aligned} \text{Using } \int (ax+b)^n \, dx &= \frac{1}{a(n+1)} (ax+b)^{n+1} + C \\ &= \frac{2}{9} (3x-1)\sqrt{3x-1} + C \end{aligned}$$

$$Q.(3)(a) 2 \sin \theta = \sqrt{3}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = \frac{\pi}{3} \text{ or } \theta = \frac{2\pi}{3}$$



(i) Midpoint AC

$$= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{-1+1}{2}, \frac{4+2}{2} \right)$$

$$= (0, 3)$$

Similarly, midpoint BD

$$= \left(\frac{-2+2}{2}, \frac{1+5}{2} \right)$$

$$= (0, 3)$$

(ii) ABCD is a parallelogram [diagonals bisect each other]

(iii) $m_{AC} = \frac{y_2 - y_1}{x_2 - x_1}$

$$= \frac{2-4}{0-(-1)}$$

$$m_{AC} = -1$$

Similarly, m_{BD}

$$= \frac{5-1}{2-(-2)}$$

$$m_{BD} = 1$$

(iv) $m_{AC} \times m_{BD}$

$$= -1 \times 1$$

$$= -1$$

Hence AC $\perp$ BD

$\therefore$ ABCD is a rhombus [diagonals $\perp$ and bisect each other]

(v) Length AC = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$= \sqrt{(-1+1)^2 + (2-4)^2}$$

$$= \sqrt{8} \text{ units}$$

Similarly, Length BD

$$= \sqrt{(2+2)^2 + (5-1)^2}$$

$$= \sqrt{32} \text{ units}$$

(vi) Area rhombus ABCD = $\frac{1}{2}xy$ [x, y diagonals]

$$= \frac{1}{2} \times \sqrt{8} \times \sqrt{32}$$

$$= 8 \text{ square units}$$

Q.(4)(a)(i) N° of bricks = $2+5+8+\dots$

$\rightarrow$ Arithmetic series: $a=2, d=3, n=20$ [for 1st 20 rows]

By $S_n = \frac{n}{2}(2a + (n-1)d)$

$$S_{20} = \frac{20}{2}(2 \times 2 + 19 \times 3)$$

$$= 610$$

$\therefore$ It would take 610 bricks to fill 20 rows.

(ii) If $S = 1000$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$1000 = \frac{n}{2}[2 \times 2 + (n-1) \times 3]$$

$$1000 = \frac{n}{2}[3n+1]$$

$$2000 = 3n^2 + n$$

$$0 = 3n^2 + n - 2000$$

Using the quadratic formula,

$$n = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1^2 - 4 \times 3 \times -2000}}{2 \times 3}$$

$$n = \frac{-1 \pm \sqrt{24001}}{6}$$

$$\therefore n = \frac{-1 + \sqrt{24001}}{6} \text{ or } n = \frac{-1 - \sqrt{24001}}{6}$$

$$n = 25.6 \dots \text{ or } n = -25.987 \dots$$

As n is a number of rows, it must be positive.

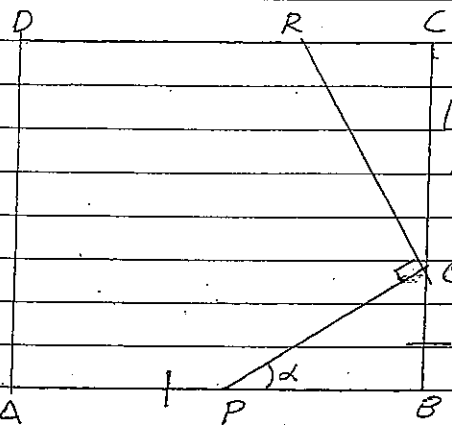
$\therefore n = 25$. [the 26th row would take 1027 bricks

$\rightarrow$ too many]

$\therefore$ 25 rows can be made [complete]

using 1000 bricks.

Q. (4)(b)

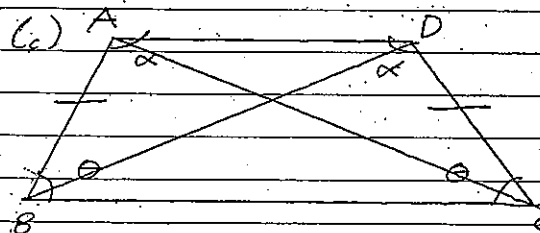


(i) $\angle PBQ = \angle QCR = 90^\circ$
 [interior $\angle$'s of square ABCD]
 Let $\angle QPB = \alpha$
 $\therefore \angle PQB = 90^\circ - \alpha$ [$\angle$ sum $\triangle PBQ$]
 $\angle RQC = \alpha$ [$\angle$'s in straight line BQC]
 $\therefore \triangle PBQ \cong \triangle QCR$ [AAS]

(ii) Hence $CR = BQ$ [matching sides, $\triangle PBQ \cong \triangle QCR$].

$CR = AP$ [data $\rightarrow BQ = AP$].

$\therefore CR = AP$



(i) In $\triangle ABC$ & $\triangle DCB$
 BC is common
 $AB = DC$ [data]
 $\angle ABC = \angle DCB$ [data]
 $\therefore \triangle ABC \cong \triangle DCB$ [SAS]

(ii) In $\triangle BAD$ and $\triangle CDA$

$BD = CA$ [matching sides in $\triangle ABC$ & $\triangle DCB$].

$BA = CD$ [data - don't need to restate this]

AD common

$\therefore \triangle BAD \cong \triangle CDA$ [SSS]

(iii) Let $\angle BAD = \alpha$

$\angle CDA = \alpha$ [matching $\angle$'s, $\triangle BAD \cong \triangle CDA$]

$\therefore \angle$ sum quadrilateral ABCD $= 2\alpha + 2\theta$

$2\alpha + 2\theta = 360^\circ$ [$\angle$ sum quadrilateral ABCD]

$\alpha + \theta = 180^\circ$

$\therefore AD \parallel BC$ [co-interior $\angle$'s supplementary] $\Rightarrow ABCD$ is a trapezium.

Q. (5)(a) $y = e^{2x}$

When $x=1$, $y = e^2$

$$\frac{dy}{dx} = 2e^{2x}$$

When $x=1$, $\frac{dy}{dx} = 2e^2$

By $y - y_1 = m(x - x_1)$

$$y - e^2 = 2e^2(x - 1)$$

$$y - e^2 = 2e^2x - 2e^2$$

$$y = 2e^2x - e^2$$

(b)(i) y intercept $= 0$

x intercepts: $y = 0$

$$3x^2 - 2x^3 = 0$$

$$x^2(3 - x) = 0$$

x intercepts $= 0$ & 3 .

(ii) $f'(x) = 6x - 3x^2$

$$1) f'(x) = 0, 6x - 3x^2 = 0$$

$$3x(2 - x) = 0$$

$$x = 0 \text{ or } x = 2$$

When $x = 0$, $f(x) = 0$ [y intercept]

$$\text{When } x = 2, f(x) = 3 \times 2^2 - 2^3 = 4$$

Nature of turning points: $f''(x) = 6 - 6x$

At $(0, 0)$ $f''(x) = 6 - 6 \times 0 = 6 \Rightarrow$ Local MINIMUM at $(0, 0)$

At $(2, 4)$ $f''(x) = 6 - 6 \times 2 = -6 \Rightarrow$ Local MAXIMUM at $(2, 4)$

(iii) Point of inflexion: $f''(x) = 0$

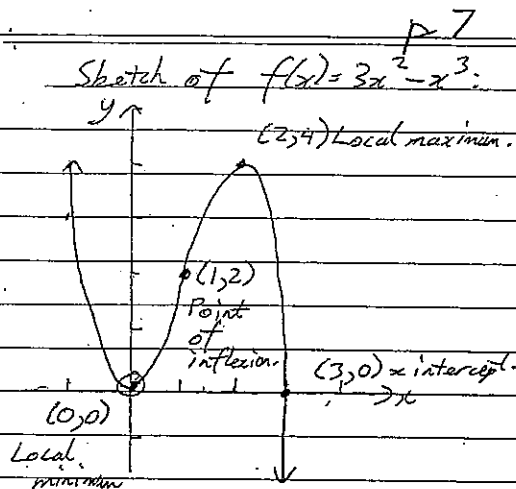
$$6 - 6x = 0$$

$$x = 1$$

$$\text{At } x = 1, f(x) = 3 \times 1^2 - 1^3 = 2$$

$f''(x)$ DOES change sign at this point as $f''(0) > 0$ and $f''(2) < 0$ [from Part (ii)]
 $\therefore (1, 2)$ is a point of inflexion.

Q. (5) (6)
[continued]



Q. (6) (a) $e^{2x} - 2e^x - 3 = 0$

Letting $a = e^x$

$$a^2 - 2a - 3 = 0$$

$$(a-3)(a+1) = 0$$

$$a = 3 \text{ or } a = -1$$

As $a = e^x$

$$e^x = 3 \text{ or } e^x = -1$$

$x = \ln 3$ For $e^x = -1$ there are no real solutions.

Q. (6) (c) $f(x) = \int 6 \cos 2x \cdot dx$

$$= 3 \sin 2x + C$$

If $f(x) = \frac{3\sqrt{2}}{2}$ When $x = \frac{\pi}{8}$

$$\frac{3\sqrt{2}}{2} = 3 \sin \frac{\pi}{4} + C$$

$$\frac{3\sqrt{2}}{2} = -\frac{3\sqrt{2}}{2} + C$$

$$3\sqrt{2} = C$$

$$\therefore f(x) = 3 \sin 2x + 3\sqrt{2}$$

p. 8

Q. (6) (b) (i) $v = t^2 - 4t$

$$a = 2t - 4$$

(ii) Minimum velocity: $a = 0$

$$2t - 4 = 0$$

$$t = 2$$

$$v \text{ at this time} = 2^2 - 4 \times 2$$

$$= -4 \text{ m/s}$$

Checking that this is a minimum:

When $t=1$, $v = 1^2 - 4 \times 1 = -3 \text{ m/s}$ } Both $> -4 \text{ m/s}$

When $t=3$, $v = 3^2 - 4 \times 3 = -3 \text{ m/s}$ } so minimum velocity at $t=2$.

$\rightarrow$ Could also argue that $v = t^2 - 4t$ is a U (concave up) parabola so only turning point is a local minimum.

(iii) $x = \int t^2 - 4t \cdot dt$

$$= \frac{1}{3}t^3 - 2t^2 + C$$

As $x = 4$ when $t = 0$, $-\frac{1}{3} \times 0^3 + C = 4 \Rightarrow C = 4$

$$\therefore x = \frac{1}{3}t^3 - 2t^2 + 4$$

(iv) Need to find when $v = 0$:

$$t^2 - 4t = 0$$

$$t(t-4) = 0$$

$$t = 0 \text{ or } t = 4$$

$\therefore$ Particle goes backwards [negative velocity] from $t=0$ to $t=4$.

When $t=4$, $x = \frac{1}{3} \times 4^3 - 2 \times 4^2 + 4$

$$= -6\frac{2}{3}$$

$\Rightarrow$ In 1st 4 seconds, particle travels from 4 to $-6\frac{2}{3} = 10\frac{2}{3} \text{ m}$.

Particle goes FORWARDS from $t=4$ to $t=5$.

When $t=5$, $x = \frac{1}{3} \times 5^3 - 2 \times 5^2 + 4$

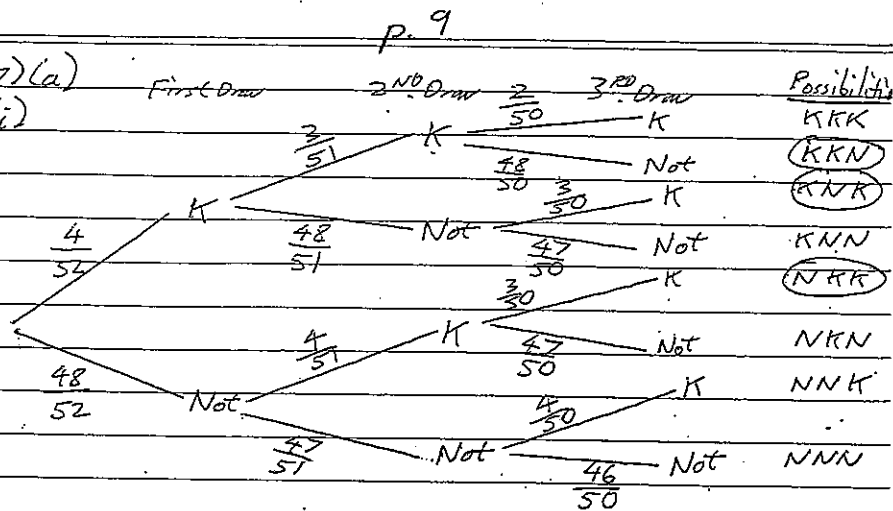
$$= -4\frac{1}{3}$$

$\Rightarrow$ In 5th second particle travels $2\frac{1}{3} \text{ m RIGHT}$.

$\therefore$ Total distance travelled = 13 m in 1st 5 seconds.

Note: Could also do $\left| \int_0^4 t^2 - 4t \cdot dt \right| + \int_4^5 t^2 - 4t \cdot dt$

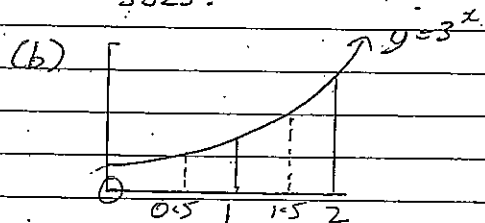
Q. (7)(a)
(i)



(ii) Probability of drawing 2 Kings in the 3 draws

$$= 3 \times \left(\frac{4}{52} \times \frac{3}{51} \times \frac{48}{50} \right)$$

$$= \frac{72}{5525}$$



$$A = \frac{1}{3} h [y_0 + 4(y_1 + y_3) + 2y_2 + y_4]$$

$$= \frac{1}{3} \times 0.5 \times [3^0 + 4(3^{0.5} + 3^{1.5}) + 2 \times 3^1 + 3^2]$$

$$= 7.7854$$

$$\approx 7.29 \text{ square units (2 DP)}$$

p. 10

Q. (7)(c)(i) When $t=0$, $P=100$
 $100 = Ae^0$
 $100 = A$

When $t=5$, $P=180$
 $180 = 100e^{5k}$
 $1.8 = e^{5k}$
 $\ln(1.8) = 5k$
 $\frac{\ln(1.8)}{5} = k \quad [= 0.11755...]$

(ii) Expected goat population in 1860: ($t=10$)
 $P = 100e^{kt}$
 $= 100e^{10 \times \frac{\ln(1.8)}{5}}$
 $= 324 \text{ goats}$

(iii) Year when $P=1000$

$1000 = 100e^{kt}$
 $10 = e^{kt}$
 $\ln(10) = kt, \quad k = \frac{\ln(1.8)}{5}$

$\therefore \frac{\ln(10)}{\left(\frac{\ln(1.8)}{5}\right)} = t$

$19.58... = t$

$\rightarrow$ During 1869 the goat population will exceed 1000.

$$(8)(a)(i) \cos x = \sqrt{3} \sin x$$

$$\div 135 \text{ by } \sqrt{3} \cos x$$

$$\frac{1}{\sqrt{3}} = \tan x$$

$$x = \frac{\pi}{6}, \frac{7\pi}{6}$$

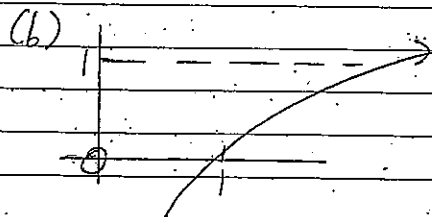
$$(ii) A = \int_{\frac{\pi}{6}}^{\frac{7\pi}{6}} \sqrt{3} \sin x - \cos x \cdot dx$$

$$= \left[-\sqrt{3} \cos x - \sin x \right]_{\frac{\pi}{6}}^{\frac{7\pi}{6}}$$

$$= \left(-\sqrt{3} \cos \frac{7\pi}{6} - \sin \frac{7\pi}{6} \right) - \left(-\sqrt{3} \cos \frac{\pi}{6} - \sin \frac{\pi}{6} \right)$$

$$= \left(\sqrt{3} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \right) - \left(-\sqrt{3} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \right)$$

$$= 4 \text{ square units}$$



$$\text{As } y = \ln(x),$$

$$x = e^y$$

$$V = \pi \int_0^1 x^2 \cdot dy$$

$$= \pi \int_0^1 e^{2y} \cdot dy$$

$$= \pi \left[\frac{1}{2} e^{2y} \right]_0^1$$

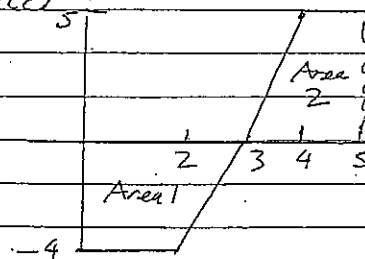
$$= \pi \left[\frac{1}{2} e^2 - \frac{1}{2} e^0 \right]$$

$$= \frac{\pi}{2} (e^2 - 1)$$

$$V = \frac{\pi}{2} (e^2 - 1) \approx 10.04 \text{ cubic units}$$

p.12

Q. (8)(c)



Distance travelled = area under graph.

$$\text{Area 1} = \frac{1}{2} h (a+b)$$

$$= \frac{1}{2} \times 4 \times (3+7)$$

$$= 10$$

→ Particle initially travels 10m backwards.

$$\text{Area 2} = \frac{1}{2} \times 5 \times (2+1)$$

$$= 7.5$$

⇒ Particle then travels

7.5m forwards.

∴ After 5 seconds, particle is at

$$2 - 10 + 7.5 = -0.5$$

= 0.5m to the LEFT of 0.

p.13

Q.(9)(a)(i) Amount in account:

Year	Start of Year	End of Year ^{x amount at start} by 1.09
1	\$8000	$\$8000 \times 1.09$
2	$\$8000 + \8000×1.09	$\$8000 \times 1.09 + \8000×1.09^2
3	$\$8000 + \$8000 \times 1.09 + \$8000 \times 1.09^2$	$\$8000 \times 1.09 + \$8000 \times 1.09^2 + \$8000 \times 1.09^3$

$\therefore$ Amount after 3 years = $\$8000(1.09 + 1.09^2 + 1.09^3)$

Using $S_n = \frac{a(r^n - 1)}{r - 1}$

$$1.09 + 1.09^2 + 1.09^3 = \frac{1.09(1.09^3 - 1)}{1.09 - 1}$$

$$= \frac{1.09(1.09^3 - 1)}{0.09}$$

$\therefore$ Total after 3 years = $\frac{\$8000 \times 1.09(1.09^3 - 1)}{0.09}$

(ii) After 20 years, Rebecca will have

$$\frac{\$8000 \times 1.09(1.09^{20} - 1)}{0.09}$$

$$= \$446\,116.24 \text{ in her account.}$$

(iii) $\frac{\$8000 \times 1.09(1.09^n - 1)}{0.09} = 1\,000\,000$

$$\begin{aligned} &\times \frac{0.09}{8000 \times 1.09} \\ &1.09^n - 1 = 10.321 \end{aligned}$$

$$1.09^n = 11.321 \dots$$

$$\ln(1.09^n) = \ln(11.321 \dots)$$

$$n \ln(1.09) = \ln(11.321 \dots)$$

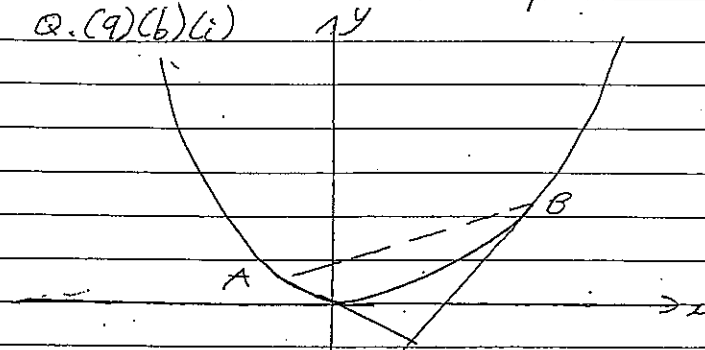
$$n = \frac{\ln(11.321 \dots)}{\ln 1.09}$$

$$n = 28.15 \dots$$

$\therefore$ It will take $28.15 = 29$ years for super to reach 1,000,000.

p.14

Q.(9)(b)(i)



By $y - y_1 = m(x - x_1)$

$$y + 2 = m(x - 3)$$

$$y + 2 = mx - 3m$$

$$0 = mx - y - 3m - 2$$

$$mx - y - 3m - 2 = 0$$

(ii) If $mx - y - 3m - 2 = 0$ is a tangent to

$x^2 = 8y$ then there is only 1 solution when $mx - y - 3m - 2 = 0$ & $x^2 = 8y$ are solved simultaneously.

$$mx - y - 3m - 2 = 0 \quad (1)$$

$$x^2 = 8y \quad (2) \text{ Sub. (2) in (1):}$$

$$mx - \frac{1}{8}x^2 - 3m - 2 = 0$$

$$8mx - x^2 - 24m - 16 = 0$$

$$x^2 - 8mx + 24m + 16 = 0 \Rightarrow \text{Quadratic equation.}$$

If a tangent then $\Delta = 0$

$$(-8m)^2 - 4 \times 1 \times (24m + 16) = 0$$

$$64m^2 - 96m - 64 = 0$$

p. 15

$$\begin{aligned} \text{Q. (9)(b)(iii)} \quad 64m^2 - 96m - 64 &= 0 \\ 2m^2 - 3m - 2 &= 0 \\ (2m+1)(m-2) &= 0 \\ m = -\frac{1}{2} \text{ or } m = 2. \end{aligned}$$

If $m = -\frac{1}{2}$, x value where

$$mx - y - 3m - 2 = 0 \& \\ x^2 = 8y$$

can be solved simultaneously.

Sub. into equation (A): from Part (ii)

$$x^2 - 8mx + 24m + 16 = 0$$

$$x^2 - 8 \times -\frac{1}{2}x + 24 \times -\frac{1}{2} + 16 = 0$$

$$x^2 + 4x + 4 = 0$$

$$(x+2)^2 = 0$$

$$x = -2$$

$$\text{As } x^2 = 8y, (-2)^2 = 8y$$

$$\frac{4}{8} = y$$

$\therefore$ Point A = $(-2, \frac{1}{2})$.

If $m = 2$, x value where

$$mx - y - 3m - 2 = 0 \&$$

$$x^2 = 8y \text{ can be solved simultaneously}$$

$$x^2 - 8mx + 24m + 16 = 0$$

$$x^2 - 8 \times 2x + 24 \times 2 + 16 = 0$$

$$x^2 - 16x + 64 = 0$$

$$(x-8)^2 = 0$$

$$x = 8 \Rightarrow \text{By } x^2 = 8y, 8^2 = 8y$$

$$8 = y$$

Point B = $(8, 8)$

p. 16

$$\text{Q. (9)(b)(iv) Chord AB:}$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{8 - \frac{1}{2}}{8 - (-2)}$$

$$= \frac{\frac{15}{2}}{10} = \frac{3}{4}$$

$$\text{By } y - y_1 = m(x - x_1)$$

Equation of chord AB:

$$y - 8 = \frac{3}{4}(x - 8)$$

y intercept of chord AB: ($x=0$)

$$y - 8 = \frac{3}{4}(0 - 8)$$

$$y - 8 = -6$$

$$y = 2$$

y intercept of chord AB = $(0, 2)$.

Focus of parabola $x^2 = 8y$

$$\text{By } x^2 = 4ay,$$

$$4a = 8 \Rightarrow a = 2$$

$\therefore$ Focus = $(0, a) = (0, 2)$

$\therefore$ Chord AB is a focal chord.

p. 17

Q. (10)(a) (i) Amount left to repay

Month	Start of Month	End
1	\$500 000	$\$500 000 \times 1.006 - P$
2		$1.006 \times (\$500 000 \times 1.006 - P) - P$ $= \$500 000 \times 1.006^2 - P(1 + 1.006)$
3		$[\$500 000 \times 1.006^2 - P(1 + 1.006)] \times 1.006 - P$ $= \$500 000 \times 1.006^3 - P(1 + 1.006 + 1.006^2)$

Note: 7.2% P.A. = 0.6% per month.

(ii) Amount left after n months

$$= \$500 000 \times 1.006^n - P(1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1})$$

$$\text{Using } S_n = \frac{a(r^n - 1)}{r - 1}$$

$$1 + 1.006 + 1.006^2 + \dots + 1.006^{n-1} = \frac{1(1.006^n - 1)}{0.006}$$

∴ Amount left to repay after n months

$$= \$500 000 \times 1.006^n - \frac{P(1.006^n - 1)}{0.006}$$

(iii) After 20 years = 240 months,

Amount left to repay = \$0

$$\$500 000 \times 1.006^{240} - \frac{P(1.006^{240} - 1)}{0.006} = 0$$

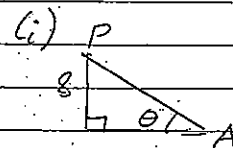
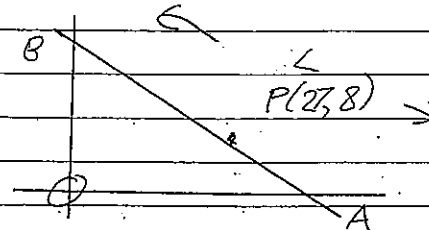
$$\$500 000 \times 1.006^{240} = \frac{P(1.006^{240} - 1)}{0.006}$$

$$\$3936.75 = P$$

→ Harika's monthly repayments amount to \$3937 (per month)

p. 18

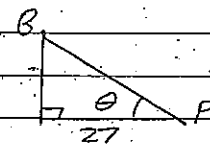
Q. (10)(b)



Distance PA:

$$\sin \theta = \frac{8}{PA}$$

$$\therefore PA = \frac{8}{\sin \theta}$$



Distance PB:

$$\cos \theta = \frac{27}{PB}$$

$$PB = \frac{27}{\cos \theta}$$

$$\therefore L = \text{Distance PA} + \text{Distance PB}$$

$$= \frac{8}{\sin \theta} + \frac{27}{\cos \theta}$$

$$(ii) \frac{dL}{d\theta} = -\frac{8}{\sin^2 \theta} \times \cos \theta - \frac{27}{\cos^2 \theta} \times -\sin \theta$$

$$= \frac{27 \sin \theta}{\cos^2 \theta} - \frac{8 \cos \theta}{\sin^2 \theta}$$

$$= \frac{27 \sin^3 \theta - 8 \cos^3 \theta}{\cos^2 \theta \cdot \sin^2 \theta}$$

(iii) L is a minimum where $\frac{dL}{d\theta} = 0$

$$\text{i.e. } \frac{27 \sin^3 \theta - 8 \cos^3 \theta}{\cos^2 \theta \cdot \sin^2 \theta} = 0$$

$$\cos^2 \theta \cdot \sin^2 \theta \times 8 \cos^3 \theta = 27 \sin^3 \theta$$

P.T.O. →

Q. (10)(b) (continued)

$$(iii) \quad 27 \sin^3 \theta - 8 \cos^3 \theta = 0$$

 $\div$ BS by $\cos^3 \theta$
 $\cos \theta \neq 0$ [must be as θ is acute]

$$27 \tan^3 \theta - 8 = 0$$

$$27 \tan^3 \theta = 8$$

$$\tan^3 \theta = \frac{8}{27}$$

$$\tan \theta = \frac{2}{3}$$

[as θ is acute, $\tan \theta$ is positive]

Checking to see if L is minimum:

If $\tan \theta = 1$ [$\theta >$ than if $\tan \theta = \frac{2}{3}$],

$$\frac{dL}{d\theta} = \frac{27 \times \frac{\sqrt{2}}{4} - 8 \times \frac{\sqrt{2}}{4}}{\frac{1}{4}}$$

$$= 19\sqrt{2} > 0$$

If $\tan \theta = \frac{1}{\sqrt{3}}$ [$\theta <$ than if $\tan \theta = \frac{2}{3}$],

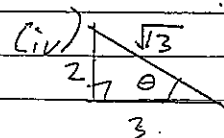
$$\frac{dL}{d\theta} = \frac{27 \times \frac{1}{8} - 8 \times \frac{3\sqrt{3}}{8}}{\frac{1}{4} \times \frac{3}{4}}$$

$$= \frac{27 - 24\sqrt{3}}{81} \times \frac{162}{31}$$

$$= 9 - 8\sqrt{3} < 0$$

$\therefore$ As $\frac{dL}{d\theta} < 0$ where $\theta < \theta$ if $\tan \theta = \frac{2}{3}$ & $\frac{dL}{d\theta} > 0$ where $\theta >$ if $\tan \theta = \frac{2}{3}$

$\Rightarrow$ Minimum where $\tan \theta = \frac{2}{3}$

(iv)  $\tan \theta = \frac{2}{3}$ $\sin \theta = \frac{2}{\sqrt{13}}$ $\cos \theta = \frac{3}{\sqrt{13}}$

$$\text{Minimum } L = \frac{8}{\sin \theta} + \frac{27}{\cos \theta}$$

$$= \frac{8 \times \sqrt{13}}{2} + \frac{27 \times \sqrt{13}}{3}$$

Minimum length $L = 13\sqrt{13}$ units